

Fahima Islam

Dhaka, Bangladesh | 23-50251-1@student.aiub.edu | fahimaislam1812@gmail.com | +880 1974 965529

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#) | [ResearchGate](#)

RESEARCH INTERESTS

AI for Healthcare, Health Informatics, Medical Image Analysis, Computer Vision, Vision-Language Models, Large Language Models, Explainable AI, Deep Learning, and Trustworthy AI.

EDUCATION

Bachelor of Science in Computer Science and Engineering

Jan 2023 – Sep 2026

American International University-Bangladesh (AIUB), Dhaka, Bangladesh

CGPA: 3.95 / 4.00

Dean's List: Fall 24 - Fall 26 (Six recognitions)

Honors: Merit-based scholarship of up to 60% toward the BSc degree

Major in Information Systems: Introduction to Data Science; Data Warehouse and Data Mining; Computer Vision and Pattern Recognition; Machine Learning.

Thesis: *LiteTC-PolypNet++: A Semi-Supervised Boundary-Refined and Temporal-Consistency Guided Lightweight Framework for Stable Real-Time Polyp Segmentation.*

Higher Secondary Certificate

2021

GPA: 5.00 / 5.00

Secondary School Certificate

2019

GPA: 5.00 / 5.00

RESEARCH EXPERIENCE

Research Assistant

April 2026 – Present

Elite Research Lab PLC, Queens, NY, USA (Remote)

Supervisor: Md Kishor Morol

- Conduct comprehensive audits of publicly available PCOS tabular and image datasets for data quality, diagnostic leakage, duplication, and shortcut learning.
- Identified criterion-feature leakage in a Kaggle PCOS dataset with 23.8K downloads at the time of the audit and contributed to a dataset verification atlas.
- Develop a multimodal deep learning pipeline for PCOS detection integrating ultrasound images with patient-matched clinical data, focusing on robust fusion and real-world validation.

RESEARCH MANUSCRIPTS

Conference Papers

1. **The Follicle Count Paradox: Measuring Diagnostic Leakage in AI-Driven PCOS Prediction.**
4th International Conference on Computing Advancements (ICCA 2026) [Accepted]
 - Audited PCOS tabular and image datasets for diagnostic leakage, duplication, and shortcut learning.
 - Identified criterion-feature leakage and contributed to the dataset verification atlas.
2. **CerebriLite-CAM: A Lightweight Interpretable Attention Network for Brain Tumor Classification from MRI.**
3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026) [Accepted]
 - Contributed to CerebriLite-CAM methodology and code combining MobileNetV2 and CBAM.
 - Implemented channel/spatial attention for glioma, meningioma, pituitary tumor, and no-tumor classification.

Code: [GitHub repository](#)
3. **DeepLabNet: Lightweight Liver Ultrasound Segmentation via Haar Frequency Fusion and Entropy Guided Boundary Refinement.**
2026 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES 2026) [Accepted]
 - Contributed to DeepLabNet methodology and code using DeepLabv3-MobileNetV3-Large.
 - Implemented Haar-frequency fusion, entropy-guided residual refinement, and boundary-distance supervision.
4. **Conflict Aware Trust-Based Aggregation for Poisoning Resistant Federated Medical Image Classification.**
IEEE i-COSTE 2026 [Accepted]
 - Contributed to Conflict-Aware Trust-Based Aggregation methodology and federated-training code.
 - Built PathMNIST experiments with IID/non-IID partitions and sign-flipping attacks.
5. **Improved Deep PACFNet: Multimodal Deep Learning for Cardiac Anomaly Detection Using Synchronized ECG-PCG Signals.**
5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026) [Accepted]
 - Contributed to Improved Deep PACFNet methodology and dual-branch Conv1D implementation.
 - Implemented Christov R-peak segmentation and ECG-PCG beat processing on PhysioNet/CinC 2016.

Code: [GitHub repository](#)

6. **Enhancing Student Engagement Using Generative AI Tools in Online Learning Environments.**

ICETOL 2026, Bremen, Germany [Accepted]

- Designed questionnaires measuring behavioral, emotional, and cognitive engagement.
- Collected and preprocessed paired responses from traditional and AI-assisted learning conditions.

Submitted and Under Review

7. **Budget-Aware Adaptive Mutation Feedback for LLM-Based Unit Test Generation: A Preliminary Empirical Study.**

ECCE 2027 [Submitted / Under Review]

- Built the Qwen2.5-Coder-3B-Instruct-AWQ-Defects4J pipeline for JUnit 4 test generation.
- Implemented Random/Adaptive Feedback, bounded repair, and separate guidance/held-out mutant pools.

8. **Near-Duplicate and Patient Leakage in MRI Brain Tumor Classification: A Leakage-Free Benchmark with CPU Deployment.**

ICEEICT 2027 [Submitted / Under Review]

- Built the benchmarking code for MobileNetV2, EfficientNet-B0, ResNet50, and VGG16.
- Implemented duplicate detection, patient-aware splits, ONNX export, and FP32/INT8 CPU benchmarking.

9. **X-RetinaPM: Explainable Deep Retinal Representations Driven by Lesion Supervision for Pathologic Myopia Analysis.**

ICEEICT 2027 [Submitted / Under Review]

- Contributed to X-RetinaPM methodology and code using the RETFound backbone.
- Implemented optic-disc/atrophy/detachment segmentation, fovea localization, and lesion-region pooling for classification.

10. **CATS-Thyroid: A Confidence-Aware Two-Stage Machine Learning Framework for Thyroid Disease Screening and Subtype Classification.**

ICCIT 2026 [Submitted / Under Review]

- Built CATS-Thyroid using LightGBM for screening and XGBoost for subtype classification.
- Implemented clinical-data preprocessing and confidence-based clinician referral.

11. **Pressure Ulcer Classification Using Strong Segmentation-Guided Feature Fusion and DINOv2.**

ICCIT 2026 [Submitted / Under Review]

- Contributed to StrongSeg-SegFusion methodology and code using CLIPSeg, SAM, and ConvNeXt-Tiny.
- Implemented raw-image/wound-ROI fusion and DINOv2-B/14 features with calibrated probability fusion.

12. **CARES-DINOv3: Counterfactual Artifact-Aware Risk Control for Selective Breast Ultrasound Segmentation.**

ICCIT 2026 [Submitted / Under Review]

- Contributed to CARES-DINOv3 methodology and code using DINOv3 ViT-S/16 and content-style adapters.
- Implemented counterfactual artifact tests, accept/adapt/abstain control, and style-only test-time adaptation.

13. **A Review on Machine Learning Approaches for Early Detection of Autosomal Dominant Polycystic Kidney Disease.**

ICCIT 2026 [Submitted / Under Review]

- Reviewed 38 studies covering ADPKD kidney/cyst segmentation, classification, and prognosis.
- Identified gaps in public datasets, screening cohorts, and external validation.

14. **SunLite: A Sub-Megaparameter Ordinal Classifier for Sunflower Growth-Stage and Health Recognition.**

NSysS 2026 [Submitted / Under Review]

- Wrote the methodology describing PP-LCNet-050, multi-scale fusion, attention pooling, and rank-consistent ordinal classification.
- Wrote the related-work analysis on sunflower growth-stage recognition and lightweight model deployment.

LEADERSHIP AND INVOLVEMENT EXPERIENCE

AIUB Computer Club, Assistant General Secretary, Public Relations

2023 – 2026

- Organized AIUB Cyber Gaming Fest 2025 and AIUB Programming Contest 2024; supported technical workshops, seminars, and industry visits, including a Brain Station 23 visit for 30 final-year students.

Augmedix Bangladesh, Campus Ambassador

2024 – 2026

- Organized Talent Hunt 2025 (AIUB × Augmedix) and coordinated industry visits for AIUB students.

NASA Space Apps Challenge & Microsoft Office Specialist Bangladesh Championship, Volunteer

2025

- Supported event logistics, participant coordination, and assistance during the hackathon and championship.

REFERENCES

Prof. Dr. Dip Nandi
Professor and Dean
American International
University-Bangladesh
dip.nandi@aiub.edu

Dr. Afroza Nahar
Professor
American International
University-Bangladesh
afroza@aiub.edu

Dr. Abhijit Bhowmik
Associate Professor
American International
University-Bangladesh
abhijit@aiub.edu